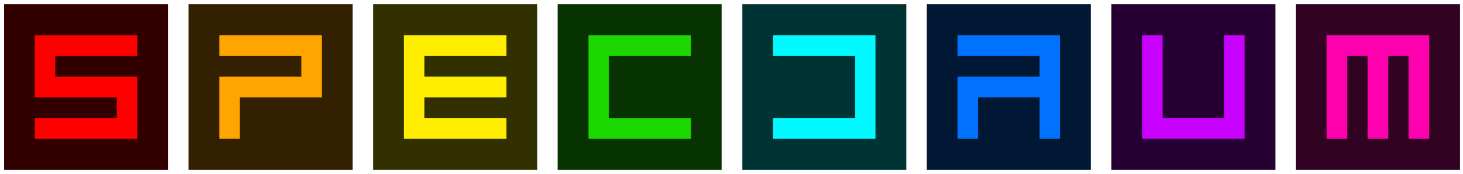
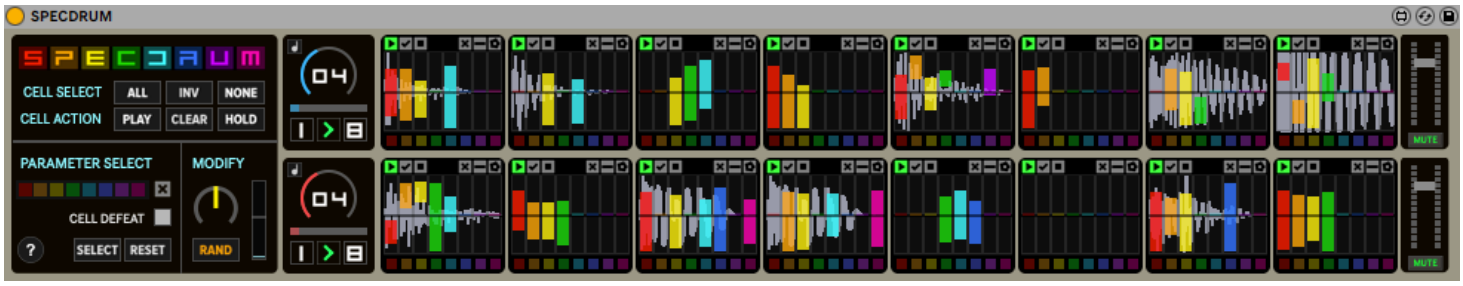


SPECDRUM by Neil Baldwin, Marmot Audio © 2023

What is SPECDRUM?



On the surface, **SPECDRUM** is a two-track sample-based drum machine. Go a little deeper and **SPECDRUM** is a randomly organic, self-catalysing sound engine. Or a slowly evolving texture generator. Or a polyrhythmic percussion section. Or a digital sprinkler that dusts your existing beats with *Hundreds and Thousands*.

SPECDRUM is driven by dual synchronised step sequencers, each with it's own independent clock-divider and playback vector. Each sequencer cell uses an audio sample as its primary sound generation to which a set of playback parameters is applied. The intensity (and sometimes character) of each playback parameter is determined by either direct setting, randomly within a defined range or just simply random.

Index

SPECDRUM Feature List
System Requirements
Installation
Loading and Saving
Interface Overview
Quick Start Guide
Reference: Playback Parameter Sliders
Reference: Playback Parameter Functions
Reference: Playback Parameter Select Buttons
Reference: Cell Playback Controls
Reference: Cell Utility Settings
Reference: Sequencer Controls
Reference: Master Controls
Reference: Master Controls - Cell Operations
Reference: Master Controls - Playback Parameter Operations
Reference: Master Controls - Slider Modification Controls
Reference: MIDI Rationale
Reference: MIDI Controls and Mapping

SPECDRUM Features

SPECDRUM is more than the sum of its parts but here's its basic parts, for parts list lovers:

- two synchronised independent 8-step sequencers, each with their own clock divider, playback modes, step start/length and swing control
- up to 16 samples can be loaded, one per sequencer step
- each step is an instance of a sample playback engine with its own individual playback parameters
- sample playback engine comprises:
    - variable playback speed
    - amplitude modulation
    - overdrive
    - resonant LPF/HPF
    - AD amplitude envelope
    - gain and panning modulation
    - variable sample offset/loop/direction
    - retriggering controls
    - self-modulation
    - final stage with low/high cut filters
- sample cell parameters can be fixed (parameter lock style) or given random ranges to behave like sample-and-hold
- multiple instances can be loaded into Live (CPU and memory allowing)
- Live preset saving including samples (saved as absolute file path references)
- Master Control section allows simultaneous and complex manipulation of sample cell parameters and playback controls

System Requirements

SPECDRUM is a "Max For Live" device therefore you need an up-to-date working installation of Ableton Live, 10 or 11, and Max For Live (which comes packaged with Live 10/11 Suite these days).

You'll need a mouse as the interface is *mouse-input-heavy*. A MIDI controller with knobs and such would also be fun for manipulation of certain controls.

Installation

The simplest way to install a new Max For Live device is to drag the .amxd file onto a track in Live. Once it has loaded, and before changing anything on the device controls, click on the Save icon in the upper-right corner of the device then accept the suggested name. It will then be saved to your Live User Library. You only need to do this once until an updated version is released.

Loading and Saving

It is possible to save SPECRDUM presets in the same way you would do for other Max For Live devices by using the Save icon in the top-right corner of the device window.

Saving will save all of the sequencer parameters and also save a *reference* to each loaded sample. **Note: If you move the source samples after saving there's no guarantee that SPECDRUM will be able to find them again.**

When reloading a saved preset, if it contains many samples it may take a couple of seconds after the